



Python Imports & Modules — Beginner Notes

1. What is an Import?

An **import** lets you use code that is written somewhere else. Instead of rewriting code, you **reuse it**.

Example:

```
</> Python  
  
import math  
  
print(math.sqrt(16))
```

Here, Python loads the built-in **math module** so you can use its functions.

2. What is a Module?

A module = a Python file

If you create a file:

```
math_utils.py
```

That file is now a module.

3. Your First Custom Module

Step 1: Create a file

math_utils.py

</> Python

```
def add(a, b):  
    return a + b  
  
def subtract(a, b):  
    return a - b
```

Step 2: Use it in another file

main.py

</> Python

```
import math_utils  
  
print(math_utils.add(5, 3))
```

✓ That's your first custom import.

4. Different Ways to Import

1. Import whole module

```
</> Python  
  
import math_utils  
  
math_utils.add(1, 2)
```

2. Import specific function

```
</> Python  
  
from math_utils import add  
  
add(1, 2)
```

3. Rename (alias)

```
</> Python  
  
import math_utils as mu  
  
mu.add(1, 2)
```

5. Why Imports Matter

Imports help you:

- Organize code
- Reuse code
- Keep files clean
- Build bigger systems

6. Making Your Own "Library"

When you have **multiple files**, group them into a folder.

Example:

```
project/  
|— main.py  
|— mylib/  
    |— __init__.py  
    |— math_utils.py  
    |— string_utils.py
```

7. What is `__init__.py`?

It's a special file that tells Python:

👉 "This folder is a package (a library)"

It can be empty:

```
</> Python  
  
# __init__.py
```

8. Importing From Your Library

```
</> Python  
  
from mylib.math_utils import add  
  
print(add(2, 3))
```

9. Cleaner Imports (Beginner Friendly Trick)

You can simplify imports.

`mylib/__init__.py`

Python

```
from .math_utils import add
```

Now in `main.py`:

Python

```
from mylib import add
```

10. Important Rules



Do:

- Use clear file names
- Keep related code together
- Use imports to avoid repetition



Avoid:

- Naming files like Python built-ins (`math.py`, `sys.py`)
- Using `from module import *`
- Putting everything in one file

11. Simple Mental Model

Think of it like this:

- File = tool
- Folder = toolbox
- Import = grabbing a tool